

Briefing Document: Green Infrastructure for Water Management (Fall 2026)

Executive Summary

The "Green Infrastructure for Water Management" course (16:180:592) at Rutgers University provides a comprehensive framework for sustainable water resource engineering, with a primary focus on urban stormwater management. The curriculum is built around the integration of **Green, Gray, Blue, and Intelligent (GGBI)** infrastructure to address the challenges of urbanization and climate change.

Key takeaways include:

- **Regulatory Compliance:** Emphasis is placed on the New Jersey Department of Environmental Protection (NJDEP) requirements, specifically N.J.A.C. 7:8, covering flood control, runoff treatment, and groundwater recharge.
- **Integrated Design:** The course moves beyond traditional "gray" infrastructure to incorporate nature-based solutions, Low Impact Development (LID), and intelligent systems across site, municipal, and regional scales.
- **Multidisciplinary Scope:** Beyond stormwater, the curriculum covers wastewater reuse, sustainable drinking water "soft-path" approaches, and energy recovery through existing water infrastructure.
- **Technical Proficiency:** Students are expected to master design procedures, cost-benefit analyses, and life-cycle assessments (LCA) using industry-standard tools such as HydroCAD and NJDEP Best Management Practices (BMPs).

1. Course Objectives and Foundational Principles

The primary objective of the course is to introduce the principles, planning, analysis, and design of green infrastructure (GI) as defined by the U.S. Environmental Protection Agency. The curriculum evaluates infrastructure alternatives at four distinct scales: site, municipal, regional, and watershed.

The GGBI Framework

Central to the course is the integrated **Green–Gray–Blue–Intelligent (GGBI)** framework:

- **Green:** Nature-based solutions and Low Impact Development (LID).
- **Gray:** Traditional engineered systems (e.g., piping, pumping, and outfalls).
- **Blue:** Water-centric elements such as ponds and wetlands.
- **Intelligent:** Integrated technology and data-driven management systems.

2. Primary Focus: Urban Stormwater Management

Stormwater management represents the core of the curriculum, addressing the impacts of urbanization and climate change on hydrology and water quality.

Regulatory and Strategic Context

- **New Jersey Requirements:** Strict adherence to NJ stormwater management requirements for flood control, groundwater recharge, and runoff treatment.
- **Urban Challenges:** Strategies for Combined Sewer Overflow (CSO) reduction and urban flood management.
- **Planning:** Integration of structural and non-structural measures into watershed and municipal planning.

Structural and Nature-Based Solutions

The course details the design and application of various Stormwater Best Management Practices (BMPs):

- **Vegetative and Soil-Based:** Rain gardens, green roofs, bioretention/biofiltration basins, and constructed wetlands.
- **Infiltration and Storage:** Pervious pavements, infiltration basins, trenches, detention basins, wet ponds, and rain barrels.
- **Engineered Systems:** Manufactured treatment devices and the hydraulic design of drainage systems, including piping and pumping.

3. Secondary Water Management Themes

While stormwater is the priority, the course synthesizes three additional pillars of sustainable water engineering:

Wastewater and Drinking Water

- **Environmental Protection:** Restoration of water environments and modeling the quality of receiving waters.
- **Regulatory Standards:** Application of Total Maximum Daily Loads (TMDLs).
- **Circular Economy:** Greywater reuse, sludge management, and beneficial use.
- **Sustainable Supply:** "Soft-path" approaches to water supply and conservation.

Energy Generation and Recovery

The curriculum explores the intersection of water infrastructure and renewable energy:

- **Anaerobic Digestion:** Energy recovery from waste.
- **Hydropower:** Utilizing existing infrastructure, such as outfalls, channels, and low-head dams, for energy generation.

4. Sustainability Tools and Environmental Impact

The course utilizes specific analytical tools to balance infrastructure benefits with environmental preservation.

Tool/Method	Application in Water Management
Life-Cycle Assessment (LCA)	Analyzing carbon emissions, energy use, and long-term costs.
Environmental Impact Assessment	Balancing the benefits of dams, canals, and pipelines against ecosystem disruption.
Benefit–Cost Analysis	Estimating costs and economic feasibility of GI alternatives.
Ecological Adaptation	Implementation of fish ladders and other measures to minimize environmental impacts.

5. Technical Resources and Assessment

Professional References and Software

The course utilizes a blend of regulatory manuals, modeling software, and academic texts:

- **NJDEP Manuals:** *New Jersey Administrative Code, N.J.A.C. 7:8* and the *NJ Stormwater Best Management Practices Manual (2026)*.
- **Software:** HydroCAD Stormwater Modeling and the Center for Neighborhood Technology's *Green Values™ Stormwater Management Calculator*.
- **Cost Estimation:** *RSMeans Data* for site work and landscape costs.

Evaluation Criteria

Academic performance is assessed through a mix of theoretical and practical applications:

- **Homework (30%) & Exam (30%):** Foundational knowledge and calculations.
- **Oral Presentation (20%):** A critical review of an instructor-approved topic (e.g., emerging GI technology or a case study).
- **Independent Project (20%):** A semester-long design or analysis project addressing a practical water management problem, requiring a comprehensive technical report.